

CREATE VPC USING OPENTOFU

Steps to create

STEP 1 – Configure AWS

- `aws configure`
- `aws sts get-caller-identity`

***Make sure the second command returns your AWS account details.**

STEP 2 – Create Project Folder

- `mkdir opentofu-vpc`
- `cd opentofu-vpc`

STEP 3 – Create OpenTofu Files

Create these 4 files:

- `main.tf`
- `variables.tf`
- `terraform.tfvars`

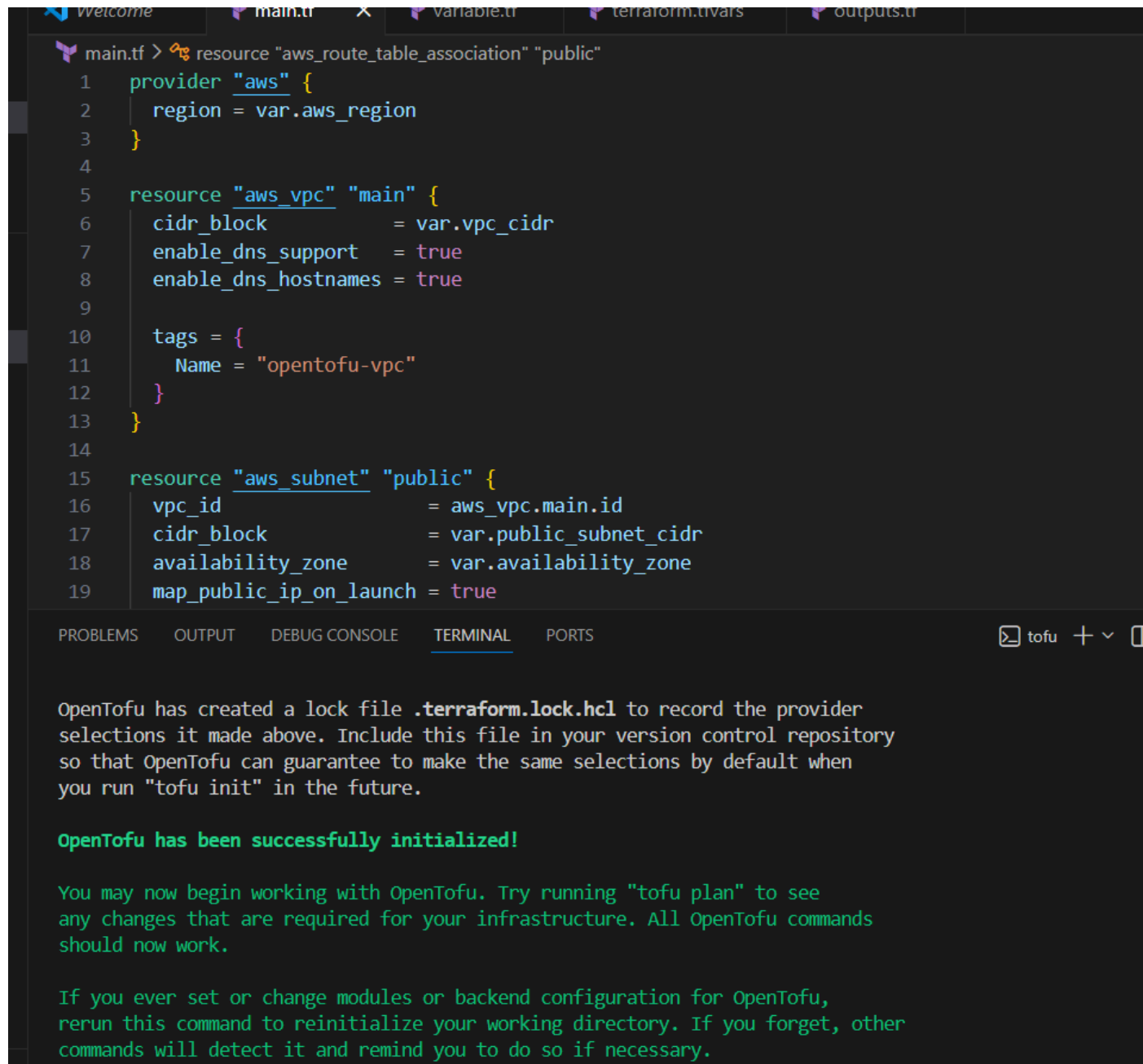
STEP 4 – Initialize OpenTofu

- `tofu init`
- *This downloads the AWS provider.**

STEP 5 – Format and Validate

- tofu fmt
- tofu validate

*If validation is successful, continue.



The screenshot shows a code editor with several tabs: `welcome`, `main.tf`, `variable.tf`, `terraform.tfvars`, and `outputs.tf`. The `main.tf` file is active, displaying the following Terraform configuration:

```
main.tf > resource "aws_route_table_association" "public"
1  provider "aws" {
2    region = var.aws_region
3  }
4
5  resource "aws_vpc" "main" {
6    cidr_block      = var.vpc_cidr
7    enable_dns_support = true
8    enable_dns_hostnames = true
9
10   tags = {
11     Name = "opentofu-vpc"
12   }
13 }
14
15 resource "aws_subnet" "public" {
16   vpc_id            = aws_vpc.main.id
17   cidr_block        = var.public_subnet_cidr
18   availability_zone = var.availability_zone
19   map_public_ip_on_launch = true
```

Below the code editor is a terminal window with the following output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
tofu + v
```

OpenTofu has created a lock file `.terraform.lock.hcl` to record the provider selections it made above. Include this file in your version control repository so that OpenTofu can guarantee to make the same selections by default when you run "tofu init" in the future.

OpenTofu has been successfully initialized!

You may now begin working with OpenTofu. Try running "tofu plan" to see any changes that are required for your infrastructure. All OpenTofu commands should now work.

If you ever set or change modules or backend configuration for OpenTofu, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

STEP 6 – Check What Will Be Created

➤ tofu plan

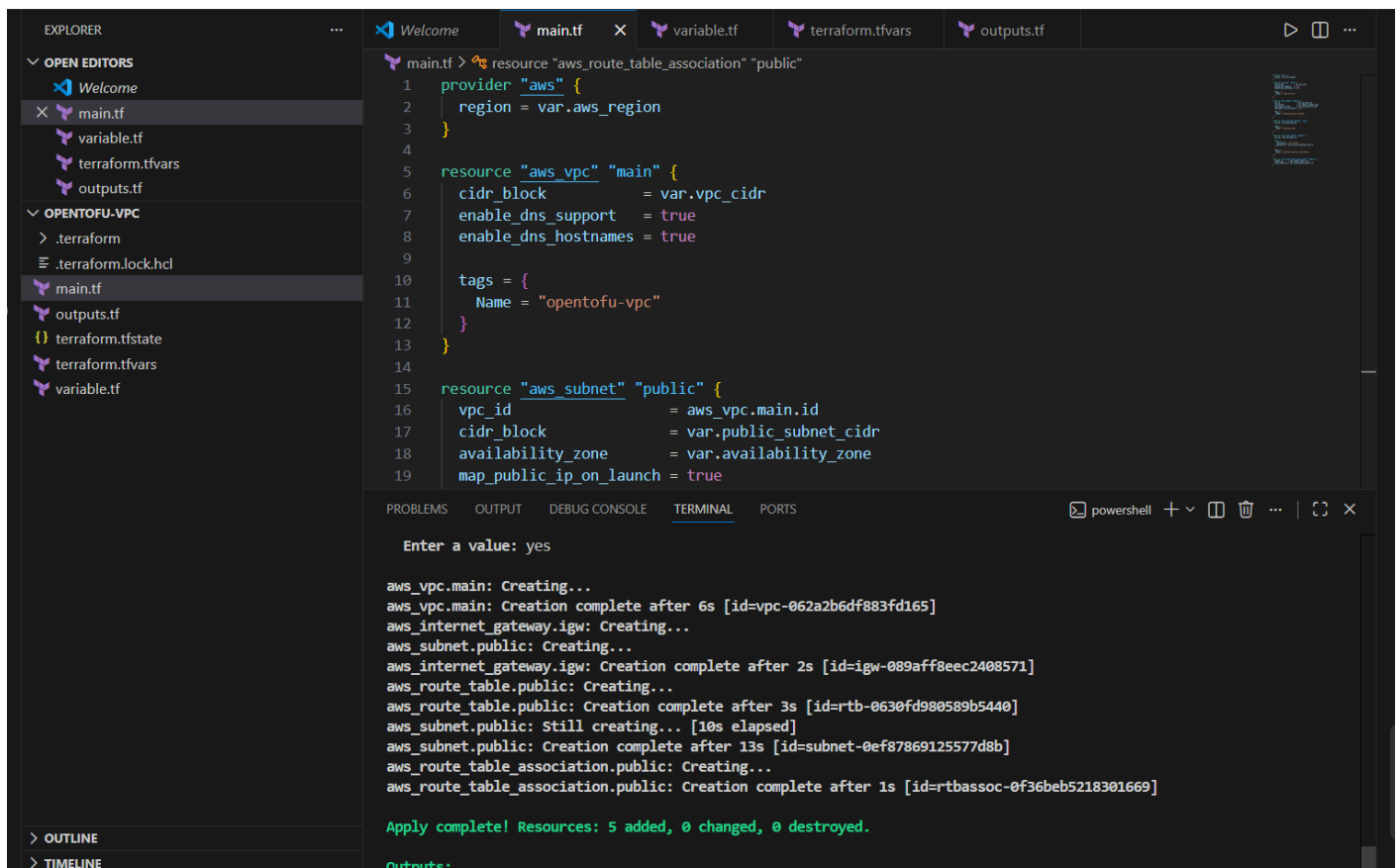
*This shows the resources OpenTofu will create.

STEP 7 – Create the VPC

➤ tofu apply

*When asked for confirmation, type:

*yes



The screenshot shows the OpenTofu workflow in VS Code. The Explorer pane on the left shows the project structure with files like `main.tf`, `variable.tf`, `terraform.tfvars`, and `outputs.tf`. The main editor displays the `main.tf` file with the following configuration:

```
1 provider "aws" {
2   region = var.aws_region
3 }
4
5 resource "aws_vpc" "main" {
6   cidr_block      = var.vpc_cidr
7   enable_dns_support = true
8   enable_dns_hostnames = true
9
10  tags = {
11    Name = "opentofu-vpc"
12  }
13 }
14
15 resource "aws_subnet" "public" {
16   vpc_id            = aws_vpc.main.id
17   cidr_block        = var.public_subnet_cidr
18   availability_zone = var.availability_zone
19   map_public_ip_on_launch = true
20 }
```

The bottom panel shows the `TERMINAL` output of the `tofu apply` command. It starts with the prompt `Enter a value: yes`. The output shows the successful creation of the VPC and its associated resources:

```
aws_vpc.main: Creating...
aws_vpc.main: Creation complete after 6s [id=vpc-062a2b6df883fd165]
aws_internet_gateway.igw: Creating...
aws_subnet.public: Creating...
aws_internet_gateway.igw: Creation complete after 2s [id=igw-089aff8eec2408571]
aws_route_table.public: Creating...
aws_route_table.public: Creation complete after 3s [id=rtb-0630fd980589b5440]
aws_subnet.public: Still creating... [10s elapsed]
aws_subnet.public: Creation complete after 13s [id=subnet-0ef87869125577d8b]
aws_route_table_association.public: Creating...
aws_route_table_association.public: Creation complete after 1s [id=rtbassoc-0f36beb5218301669]

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.
```

The `Outputs:` section is visible at the bottom of the terminal panel.

STEP 8 – Check Created Resources

- tofu output
- Then open:

*AWS Console → VPC

*Verify:

*VPC

Your VPCs

VPCs | VPC encryption controls

Your VPCs (1/2) [Info](#)

Last updated 25 minutes ago [Actions](#) [Create VPC](#)

Find VPCs by attribute or tag

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	
opentofu-vpc	vpc-062a2b6df883fd165	Available	-	-	Off	1

vpc-062a2b6df883fd165 / opentofu-vpc

Details | Resource map | CIDRs | Flow logs | Tags | Related resources | Integrations

Details

VPC ID vpc-062a2b6df883fd165	State Available	Block Public Access Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-02d9c5869642bdacc	Main route table rtb-08e0e85615f44323b
Main network ACL	Default VPC	IPv4 CIDR	IPv6 CIDR

*Public Subnet

Subnets (1/7) Info Last updated 25 minutes ago Actions Create subnet

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	-		Available		Off	172.31.64.0
☐	-		Available		Off	172.31.0.0/
☒	opentofu-public-subnet	subnet-0ef87869125577d8b	Available	vpc-062a2b6df883fd165 open...	Off	10.0.1.0/24

subnet-0ef87869125577d8b / opentofu-public-subnet

[Details](#) [Flow logs](#) [Route table](#) [Network ACL](#) [CIDR reservations](#) [Sharing](#) [Tags](#) [Related resources](#)

Details

Subnet ID subnet-0ef87869125577d8b	Subnet ARN arn:aws:ec2:us-east-1:094005226683:subnet/subnet-0ef87869125577d8b	State Available	Block Public Access Off
IPv4 CIDR 10.0.1.0/24	Available IPv4 addresses 251	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone		VPC	Route table

*Internet Gateway

Internet gateways (1/2) Info Last updated 25 minutes ago Actions Create internet gateway

Find internet gateways by attribute or tag

	Name	Internet gateway ID	State	VPC ID	Owner
☒	opentofu-igw	igw-089aff8eec2408571	Attached	vpc-062a2b6df883fd165 opentofu-vpc	
☐	-		Attached		

igw-089aff8eec2408571 / opentofu-igw

[Details](#) [Tags](#)

Details

Internet gateway ID igw-089aff8eec2408571	State Attached	VPC ID vpc-062a2b6df883fd165 opentofu-vpc	Owner
---	--------------------------	---	------------------

*Route Table

Route tables (1/3) [Info](#) Last updated 2 minutes ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-				Yes	
☒	opentofu-public-route-table	rtb-0630fd980589b5440	subnet-0ef87869125577...	-	No	vpc-062a2b6df883fd165 opentofu-vpc
☐	-				Yes	

rtb-0630fd980589b5440 / opentofu-public-route-table

[Details](#) [Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#) [Related resources](#)

Details

Route table ID
[rtb-0630fd980589b5440](#)

VPC
[vpc-062a2b6df883fd165](#) | [opentofu-vpc](#)

Main
No

Owner ID
[\[redacted\]](#)

Explicit subnet associations
[subnet-0ef87869125577d8b](#) / [opentofu-public-subnet](#)

Edge associations
-

STEP 9 – Delete Resources After Practice

- tofu destroy
- Type: yes

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Mahin\opentofu-vpc> tofu destroy
aws_vpc.main: Refreshing state... [id=vpc-062a2b6df883fd165]
aws_internet_gateway.igw: Refreshing state... [id=igw-089aff8eec2408571]
aws_subnet.public: Refreshing state... [id=subnet-0ef87869125577d8b]
aws_route_table.public: Refreshing state... [id=rtb-0630fd980589b5440]
aws_route_table_association.public: Refreshing state... [id=rtbassoc-0f36beb5218301669]

OpenTofu used the selected providers to generate the following execution plan. Resource actions are indicated with
the following symbols:
  - destroy

OpenTofu will perform the following actions:

# aws_internet_gateway.igw will be destroyed
- resource "aws_internet_gateway" "igw" {
  - arn      = "arn:aws:ec2:us-east-1:094005226683:internet-gateway/igw-089aff8eec2408571" -> null
  - id      = "igw-089aff8eec2408571" -> null

```

Enter a value: yes

```
aws_route_table_association.public: Destroying... [id=rtbassoc-0f36beb5218301669]
aws_route_table_association.public: Destruction complete after 2s
aws_subnet.public: Destroying... [id=subnet-0ef87869125577d8b]
aws_route_table.public: Destroying... [id=rtb-0630fd980589b5440]
aws_subnet.public: Destruction complete after 2s
aws_route_table.public: Destruction complete after 3s
aws_internet_gateway.igw: Destroying... [id=igw-089aff8eec2408571]
aws_internet_gateway.igw: Destruction complete after 1s
aws_vpc.main: Destroying... [id=vpc-062a2b6df883fd165]
aws_vpc.main: Destruction complete after 1s
```

Destroy complete! Resources: 5 destroyed.

PS C:\Users\Mahin\opentofu-vpc> █

FINAL FLOW

AWS Configure



Create Project



Create .tf Files



Paste Code



tofu init



tofu validate



tofu plan



tofu apply



Verify in AWS



tofu destroy